

Joel Rogers

Mechatronics Engineering Portfolio

UAV V2



Fabricated an ultra-strong **fibre glass** core section (light grey) for strength, small wall thickness and weight.



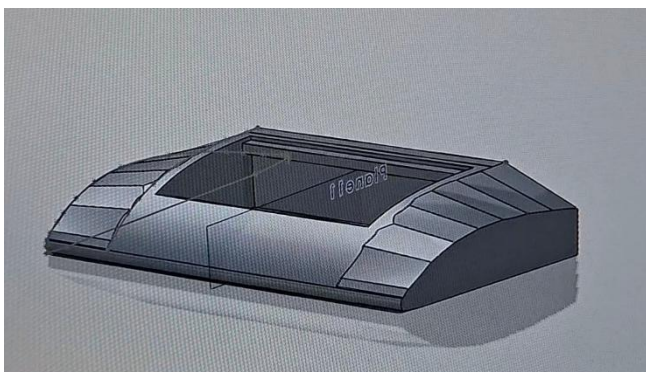
Designed a **flight controller (KiCAD)** with integrated GPS, RF receiver and IMU module for autonomous flight and fly by wire pilot assistance.



Hand launch of the UAV V2.



A display of the **fly by wire assistance** where the elevons are attempting to relevel the plane.



Designed, modelled (**SolidWorks**), manufactured and tested a flying wing UAV platform.



Design changes through the learning of the prototype (**design through iteration**).

UAV Prototype



Designed, modelled (**SolidWorks**), manufactured and tested a flying wing UAV platform.



Integrated long range **RF** communication for **command and control**.



Achieved flight after six failed attempts and subsequent redesigns.

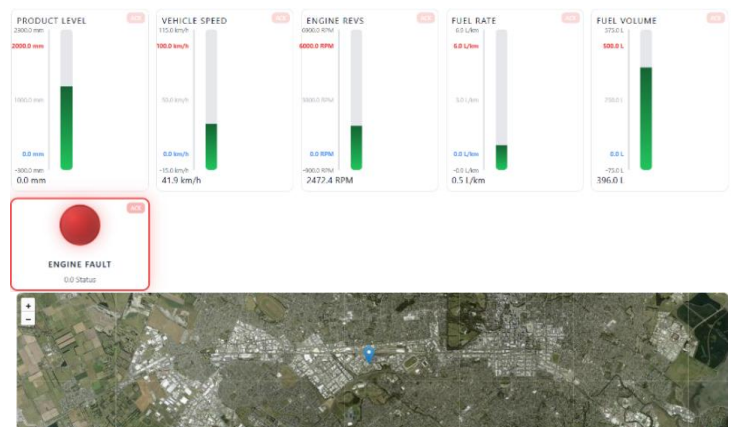


Designing through failure and partial destruction (**Design through iteration**).

Sentinel Wireless IoT Module

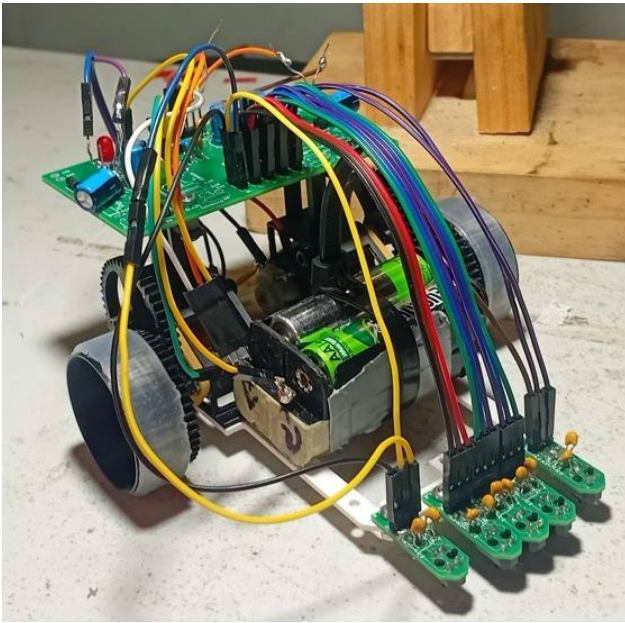


Multiprotocol wireless communicating IoT ESP32 gateway, communicating through **WiFi**, **4G**, **Ethernet** and obtaining **GPS** telemetry.

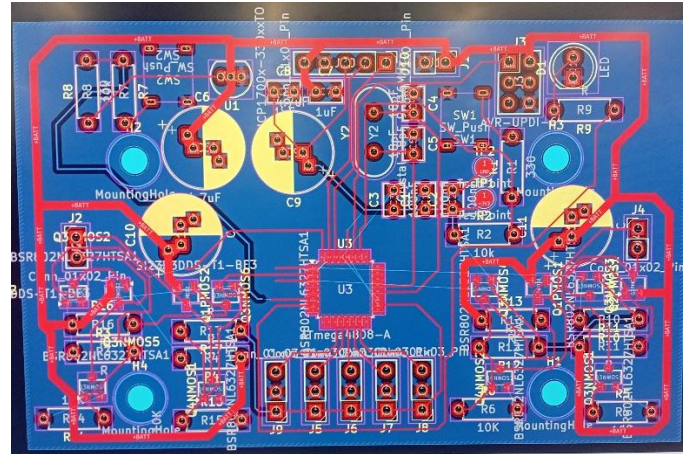


Also included the construction of a **server**, **database** and **web UI**.

Line Following Robot



Broke the record for the annual University of Canterbury line following robot competition, with our fastest time being **5.81s**. Designed, 3D modelled (**Fusion 360/SolidWorks**), **3D printed** and tested numerous gearing systems, motor mounts and chassis.



Designed (**KiCAD**) and assembled the main **microcontroller PCB** and smaller sensor PCBs. Programmed the robot in **Arduino IDE** and integrated an ESP8266 for **wireless communication** and PID value changes over **WIFI**.